ISIT219 Knowledge and Information Engineering

Assignment 2 Part A and B
Group Assignment

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Assignment 2 Part A – Data Minning

1. Introduction

This report compares multiple analytics tools, documents the features actually used, justifies three knowledge-creation techniques (classification, clustering, association), explains sources and data cleaning, presents two patterns per source, addresses inconsistencies, and provides actionable recommendations. Methods and software are cited at first mention to support claims about technique choice, validation and interpretation (Agrawal & Srikant, 1994; Breiman, 2001; Demšar et al., 2013; Harris et al., 2020; Hunter, 2007; Lloyd, 1982; MacQueen, 1967; McKinney, 2010; Pedregosa et al., 2011; Rousseeuw, 1987; Thorndike, 1953).

2. Tool comparison and selection

We used a hybrid approach combining RapidMiner Studio, Google Colab (Python stack), and Orange. This satisfies the requirement to compare at least two analytics tools and to discuss features used. RapidMiner and Orange provided visual, auditable workflows; Colab provided code-first flexibility via the standard scientific Python stack (McKinney, 2010; Harris et al., 2020; Pedregosa et al., 2011; Hunter, 2007; Demšar et al., 2013).
RapidMiner Studio. RapidMiner offers a visual workflow with drag-and-drop operators and typed ports that prevent invalid connections, plus built-in learners such as Decision Tree and k-means and repository utilities that help document repeatable pipelines. Its visual operators for Sort and Store helped capture lineage and reproducibility (RapidMiner, Inc., n.d.; Lloyd, 1982; MacQueen, 1967).
Google Colab (Python). Colab was used with pandas, NumPy, scikit-learn and matplotlib for data preparation, modeling and charts (McKinney, 2010; Harris et al., 2020; Pedregosa et al., 2011; Hunter, 2007). We implemented Apriori for market-basket mining and generated publication-quality visuals. Typical parameters were minimum support ≈ 0.01 and minimum confidence ≈ 0.20, with rules ranked by lift and itemset length capped at 3 (Agrawal & Srikant, 1994; Brin et al., 1997).
Orange. Orange's visual workflows were used for role assignment, imputation, one-hot encoding, normalization, evaluation with 10-fold stratified cross-validation, and clustering analysis with Silhouette validation and Box-plot summaries, enabling rigorous and reproducible analysis (Demšar et al., 2013; Kohavi, 1995; Rousseeuw, 1987).
Selection outcome. We paired RapidMiner and Orange for visual-first classification and clustering, and Colab/Python for transactional wrangling and rule mining (Agrawal & Srikant, 1994; Demšar et al., 2013; Pedregosa et al., 2011).

3. Data analytics tools and features utilized

RapidMiner. Repository retrieval and storage; data cleaning via Replace Missing Values and Filter Examples; Decision Tree with pruning for transparent splits; k-means with centroid inspection plus elbow checks to select candidate k values; Sort and Store for

documenting results (Breiman et al., 1984; Lloyd, 1982; MacQueen, 1967; Thorndike, 1953).
Google Colab (Python). pandas/NumPy to trim strings, convert blanks appropriately, and create both long and one-hot tables; Apriori to mine frequent itemsets and rules (support, confidence, lift); matplotlib for co-occurrence heatmaps and support-vs-confidence charts. Parameters were min_support 1%, min_confidence 20%, max_len 3 (Agrawal & Srikant, 1994; Brin et al., 1997; Harris et al., 2020; Hunter, 2007; McKinney, 2010).
Orange. File and Select Columns to define roles; Impute for missing values; Preprocess for one-hot encoding and normalization; Test & Score with 10-fold stratified CV; k-means, Silhouette Plot to select k, and Box Plot to compare features across clusters (Demšar et al., 2013; Kohavi, 1995; Rousseeuw, 1987).

4. Knowledge-creation techniques and justification

The brief requires classification, clustering and association analysis; we employed all three and aligned each to the most suitable data source.
Clustering (k-means) on Survey data. We used k-means with k = 2, k-means++ initialization and at least 10 restarts, validating the choice of k by the highest average Silhouette score. k-means is efficient for Euclidean, convex clusters; k-means++ improves seeding stability; Silhouette quantifies cohesion and separation (Arthur & Vassilvitskii, 2007; Lloyd, 1982; MacQueen, 1967; Rousseeuw, 1987).
Classification (Decision Tree, Random Forest) on Profile data. We predicted Credit_Payment from demographics. Decision Trees provide interpretable decision rules; Random Forest serves as a higher-capacity, bagged ensemble baseline with built-in variance reduction (Breiman et al., 1984; Breiman, 2001).
Association analysis (Apriori) on Transactions. Association rules suit set-valued baskets and surface co-purchases that inform coupon pairing, cross-sell bundles and adjacency planning. We filtered by support and confidence and ranked by lift to emphasize deviations from independence (Agrawal & Srikant, 1994; Brin et al., 1997).

5. Information sources and data cleaning

Transactions (CSV). We trimmed item strings, deduplicated within baskets, created long and one-hot tables, and treated transactional blanks as absent (not imputed) to preserve valid support calculations. Apriori mining used support ≥ 1%, confidence ≥ 20% and max_len = 3 (Agrawal & Srikant, 1994).
Profile (XLSX). We set Credit_Payment as target, assigned numeric and categorical roles, removed rows where the target was “unknown,” imputed Age with mean and categoricals with mode, encoded categoricals (one-hot), and standardized numeric features for modeling. Evaluation used 10-fold stratified CV and AUC-based comparisons (Kohavi, 1995; Bradley, 1997; Pedregosa et al., 2011).
Survey (XLSX). The raw survey file exhibited impossible magnitudes on a 0–10 scale, indicating corruption. Orange preprocessing and normalization stabilized the data. We applied min-max scaling to [0,1] to equalize feature influence on Euclidean distances in k-means (Demšar et al., 2013; Lloyd, 1982).

6. Generated knowledge and patterns

Transactions (association rules).
Dessert add-on. A high-lift rule of the form {berries} ⇒ {whipped/sour cream} revealed a strong dessert pairing beyond baseline frequencies (Agrawal & Srikant, 1994; Brin et al., 1997).

Meal-build bundles. Patterns such as {rolls/buns, soda} ⇒ {sausage} indicated a grill/picnic triad; produce-completion rules like {other vegetables, tropical fruit} ⇒ {root vegetables} reflected "complete the meal" behavior. Whole milk appeared as a frequent companion in co-occurrence plots. High-confidence rules anchored by very popular items can still exhibit low lift, which limits promotional precision (Brin et al., 1997).

Profile (classification).
Weak predictive power of demographics. Logistic Regression, Random Forest and Decision Tree showed modest test performance, with best Test AUC ≈ 0.665 for Random Forest under cross-validation, indicating demographic fields alone are weak predictors of Credit_Payment (Bradley, 1997; Pedregosa et al., 2011).

Overfitting risk. A gap between training AUC (≈ 0.822 for Random Forest) and Test AUC (≈ 0.665) indicates overfitting and limited generalization from static demographics (Breiman, 2001; Kohavi, 1995).

Qualitative splits. Decision-tree splits suggested higher card usage among working-age, married customers with stable occupations; families showed higher spend frequency, consistent with mid-week family-oriented promotions (Breiman et al., 1984).

Survey (clustering).
Segment sizes and cohesion. The k = 2 solution, chosen by the highest average Silhouette score (≈ 0.47), produced clusters of roughly 75% and 25%, with Box-plot diagnostics consistent with robust separation under Euclidean distance (Rousseeuw, 1987; Lloyd, 1982).

Comfort as the differentiator. Product-quality, Deli-quality and Service-quality did not differ significantly between clusters ($p > 0.05$). Shopping_comfort was significantly different ($p < 0.001$) and became the basis of the segmentation.

7. Analysis of inconsistencies

Analytical disconnect (Profile vs Transactions). Demographics poorly predict Credit_Payment (modest AUC), while transactional data yields clear, actionable patterns via lift-ranked rules. Behavioral features are more predictive than static demographics (Bradley, 1997; Brin et al., 1997).

Rule strength vs item popularity. Rules involving very popular items can score high on confidence but low on lift, reducing their value for targeted promotions (Brin et al.,

1997).

Say-do gap (survey vs baskets). Customers may report quality preferences yet buy convenience add-ons (e.g., buns, soda), a common discrepancy between stated and revealed preferences.

8. Recommendations

Immediate operational focus: address universal dissatisfaction. Improve product, deli and service quality across the board before fine segmentation tactics, since both clusters showed similar dissatisfaction on core dimensions.

Predict with behavior, not demographics. Re-platform risk and propensity models on transactional features (frequency, basket composition) and on the comfort segment label; avoid reliance on static demographics for Credit_Payment prediction (Bradley, 1997; Pedregosa et al., 2011).

Target the low-comfort minority (~25%). Improve wayfinding, signage and queue experience (e.g., express lanes) to shift the low-comfort segment toward the majority.

Implement tactical bundles and adjacency. Launch a "Berry & Cream Weekend" bundle and co-locate these items; run a seasonal grill/picnic end-cap featuring sausage, buns and soda with a single coupon (Agrawal & Srikant, 1994; Brin et al., 1997).

Governance and refresh. Publish RapidMiner and Orange outputs to a dashboard and run the association-rule miner on a weekly cadence, tuning thresholds to category velocity to enable Operations to monitor lift, segment sizes, and changes over time (Demšar et al., 2013).

9. Validation settings and parameters

Association rules. Initial settings: min_support ≈ 0.01, min_confidence ≈ 0.20–0.50, max itemset length 3, rules ranked by lift (Agrawal & Srikant, 1994; Brin et al., 1997).

Classification. 10-fold stratified cross-validation for robust estimates; tree pruning to mitigate overfitting; forest depth/leaf controls for bias-variance balance; AUC used for model comparison (Breiman, 2001; Breiman et al., 1984; Bradley, 1997; Kohavi, 1995; Pedregosa et al., 2011).

Clustering. Min-max normalization to [0,1]; k = 2 selected by highest average Silhouette score (≈ 0.47); k-means++ initialization with multiple restarts (Arthur & Vassilvitskii, 2007; Rousseeuw, 1987).

10. Limitations and risks

Survey data integrity. The raw survey file contained corrupted magnitudes on a 0–10 scale; although Orange preprocessing mitigated this, source integrity remains a concern and should be addressed in future collection and validation (Demšar et al., 2013).

"Popular item" trap in rules. High-frequency items inflate confidence without improving lift, which can mislead promotion design if not screened by lift and margin constraints (Brin et al., 1997).

Model ceiling on demographics. The training–test AUC gap signals overfitting when using demographics alone; further gains likely require behavioral inputs (Bradley, 1997).

11. Next steps and measurement

Pilot cadence. Tie the comfort fixes and bundles to matched-store A/B tests with weekly dashboard refresh and rule/segment monitoring (Demšar et al., 2013).

KPIs. Track promotional conversion, attachment rate for bundle components, average basket size, category margin, queue abandonment and segment share shifts. These match the operational uses implied by association rules, clustering segments and validation practice (Agrawal & Srikant, 1994; Brin et al., 1997; Rousseeuw, 1987).

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Assignment 2 Part B Expert System

1. Introduction and Background

This research focuses on the technical support areas in General Services Administration (GSA) Education Company. The company operates a helpdesk to solve device issues faced by students or schools. These devices are installed with Device Management Application(DMA) to ensure the proper functionality and efficiency.

GSA Education provides technical support and Personal Learning Devices (PLDs) for schools and students who use iPads and Chromebooks under DMA. Mobile Guardian (MG) software controls these devices, placing child restrictions on them. It acts as a firewall, blocking inappropriate websites, manages classroom activity and locks devices at night for adequate rest. (*Designed for Schools. Loved by Parents - Mobile Guardian*, n.d.)

However, the technical support process is complex, involving multiple platforms serving different purposes. These platforms include MG, Zoho Desk, Slack & Zendesk, GSA Command Centre and Social Media (inclusive of messaging platforms).

The current technical support model for a Large-Scale Educational Device Deployment is highly manual and reactive. It relies on human expertise to diagnose and resolve occurring issues using a large set of tools creating significant operational inefficiencies.

For example, if at the start of the lesson, half the class's devices are locked or have incorrect settings, GSA Support would be contacted. The technician must then manually login to MG, diagnose the issue across multiple platforms - MG, GSA Command Centre, Slack etc to further contact the experts to fix the issue. (*EdTech | Singapore | GSA Education*, n.d.)

Based on the example, it is clear that the process is very slow and complex and relies on multiple platforms, leading to a very confusing escalation pathway. This process is also extremely inconsistent, with the same issues, having different fixes. Despite this, the different platforms allow for experts with different levels of expertise to look into the issue.

The Expert System created targets technicians, aiming to eliminate technical issues that require manual escalation to at least 70% by providing quick resolution of issues to increase the efficiency of the system.

To elaborate on our goal, we wish to resolve the weakness of the current MG through the use of expert systems which would provide quick results to the user. This ensures all users receive consistent access to the same information through a single, seamless interface, without relying on human interaction.

The Expert System would take in input such as type of device, role of user and description of the problem. Based on this information, the system would create a ticket. If there is sufficient information from past cases, the issue would be resolved and a confirmation email would be sent to the user. However, if further escalation is needed, the user would be sent an email and once the issue is resolved, the new case is then added into the expert system, expanding it.

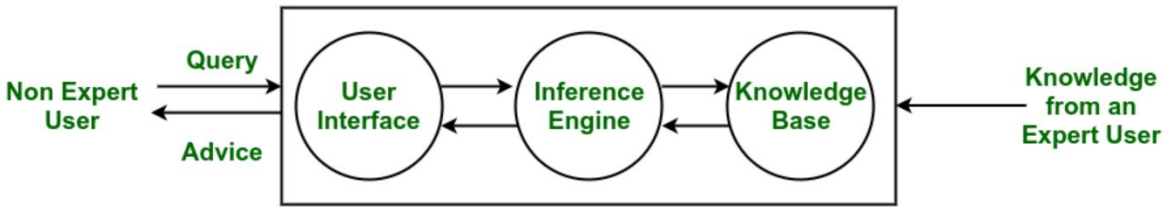
The new expert system is scalable, allowing new devices, platforms, issues to be added without reconstruction of the system. Although each domain is separate, they are linked by common working memory and inference engines. Additionally, the new system enhances data security with the reduction of manual logins to multiple administrative sites, minimising human exposure to sensitive data. All diagnostic interactions would occur in a controlled environment with predefined rulesets and access policies. This would ensure only authorised roles can access restricted portions of the Knowledge Base, preventing credential leaks or accidental data sharing between different platforms. Security measures such as audit logs are also maintained, providing a traceable record of every automated action or escalation, if any issue were to arise.

2. Methods and Tools

This section focuses on how our research and process went about along with the tools used. Through BeiRui's internship at GSA, we had brainstorming sessions and constructed the technical support workflow, focusing on the quick resolution of issues to increase efficiency of the system without human interventions. This process involved identification and classification of common issues and breaking them down into their diagnostic steps, identifying key questions a technician would ask and decision points that would lead to the solution or escalation. This process then provided us with a logical flow to our knowledge system.

Our Expert system uses case-based reasoning to diagnose and troubleshoot the common issues while adding new experiences as data in the system. To do this, we first compiled common device issues that are faced by students through research and a set of interview questions. Questions asked included their experience with MG, common device issues they run into and how long does it take to resolve each issue. (Reddy & Fields, 2022)

After which, we developed the system through a set of IF-THEN rules to go about the decision-making process. An example would be "IF there is a slow performance THEN scan the device for malware". This approach allows for case matching, comparing user input from new cases and comparing information to past cases to resolve it. (Reddy & Fields, 2022)



Case Matching is possible as the Knowledge Base, Inference Engine and User Interface all dynamically exchange knowledge within the Expert System. When a new case arrives, the information inputted by the User Interface is compared to the pre-existing rules in the Knowledge Base in the Inference Engine. Any matches would lead to the resolution of the case.

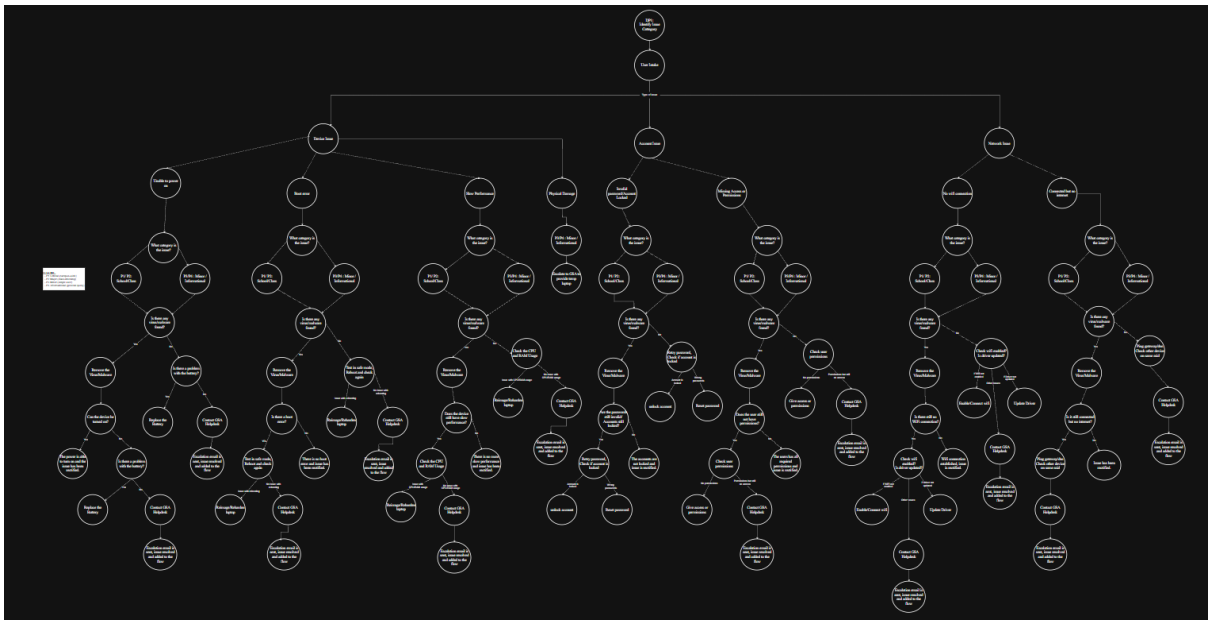
Any new cases added through the User Interface would be processed through the inference engine, creating new rules or improving old ones. These rules and outcomes of the case would be added to the Knowledge Base. Knowledge is expanded and reused across several

levels of reasoning due to internal feedback loop (identification of issue → impact assessment → decision routing).

The system's Inference Engine acts as an internal record of the reasoning path, logging fired rules and storing them in the Knowledge Base and generating conclusions. The logs fired aids the developmental team to identify any contradictory rules, enhancing the Expert System's overall knowledge quality and consistency.

The system is also designed to output the conclusion in two different ways - through the user and through the system. It shares information to the users (technicians) through the automation of the solution or in the instance that is not possible, it provides them with a step-by-step path to troubleshoot the issue. This process is immediate, providing users with feedback on the issue. Another way information is shared is through the system output through the ticket generated, there would be a payload with information such as the user metadata, the problem description and different steps used to resolve the data and which cases the expert system referenced in JSON/XML format. (GeeksforGeeks, 2025)

To operationalize the logical flow of the troubleshooting process, a Knowledge Tree was constructed using Draw.io. This visual representation captures the hierarchical structure of issues, branching from high-level categories (Device, Account, Network) into subcategories based on severity and diagnostic checks. The Knowledge Tree served as a blueprint for the system's decision-making logic and guided the development of production rules.



Based on the Knowledge Tree, production rules were authored in ES Builder to operationalize the decision flow. Each rule includes a unique Rule ID, the relevant attribute conditions such as device issue type, category level, malware presence, and diagnostic results, as well as the recommended action or resolution. Where appropriate, rules also specify whether the case should be escalated to GSA Helpdesk. These production rules allow the system to reason about new cases dynamically and guide technicians step-by-step through troubleshooting. Representative examples include rules for battery replacement when a device will not power on, escalation to the helpdesk for slow performance, granting missing account permissions, and updating network drivers for connectivity issues.

Device Issue - Battery Replacement

IF attribute device Issue
AND what is wrong with the device? unable to power on
AND what category level is the issue? p1/P2: School/Class
AND is there any virus/malware found? no
AND is there a problem with the battery? yes
THEN the Resolution is
Replace the battery.

Battery is degraded causing no power for the device.

Device Issue - Escalation

IF attribute device Issue
AND what is wrong with the device? slow performance
AND what category level is the issue? p1/P2: School/Class
AND is there any virus/malware found? no
AND check the CPU and RAM Usage no issue with CPU/RAM usage
THEN the Resolution is
Contact GSA Helpdesk for escalation.

Escalation to GSA helpdesk required for further off-site troubleshooting.

Account Issue - Grant Permissions

IF attribute account Issue
AND what is the issue with the account missing Access or Permissions
AND what category level is the issue? p1/P2: School/Class
AND is there any virus/malware found? no
AND check user permissions no valid permissions
THEN the Resolution is
Grant access or permissions.

Relevant access and permission to be granted to resolve access issue.

Network Issue - Update Driver

```
IF attribute network Issue
AND what is the issue with the network? no WiFi connection
AND what category level is the issue? p3/P4: Minor/Informational
AND is there any virus/malware found? no
AND check wifi enabled? Is driver updated? driver is not updated
THEN the Resolution is
Update driver.
```

```
Update WiFi driver to enable reconnection.
```

3. Evaluation Method

To evaluate the expert system's logical soundness, operational accuracy, and potential impact in reducing manual escalations by up to 70%, a scenario-based walkthrough was conducted alongside rule validation and output consistency checks. This was done through a scenario-based walk-through along with Rule Validation and Output Consistency checks. The walk-through enabled the team to see if the rules were operated as anticipated, if the reasoning flow is logical and if identical input patterns would return the same results. Doing so, allowed us to simulate real-world IT support instances. This guaranteed our domain accuracy and deterministic behaviour.

Our walk through involved the step-by-step simulation of the system using different test scenarios. The test cases used were first categorized by issue type - Device, Account, Network issues. Each test case consisted of symptom descriptions, relevant diagnostic results and expected resolution based on expert knowledge. This was evaluated through the correctness, efficiency, escalation appropriateness and coverage of the system. Correctness would determine if the resolution suggested by the system matched expected outcome of the test case, efficiency measures the number of steps needed to reach the resolution while escalation appropriateness refers to cases that required external (GSA Helpdesk) support, while coverage calculates the percentage of test cases successfully addressed.

However, we encountered some practical difficulties during evaluation. Ambiguous user inputs might lead to a match in multiple conditions and this requires a more careful mapping of the attribute structure. There was also a need to ensure all multi-conditional scenarios were adequately addressed by the rules, avoiding gaps or conflicts in the decision tree. Additionally, as this is based on a test case scenario, in the real-world there is more variability and complexity of issues which have not been tested or addressed.

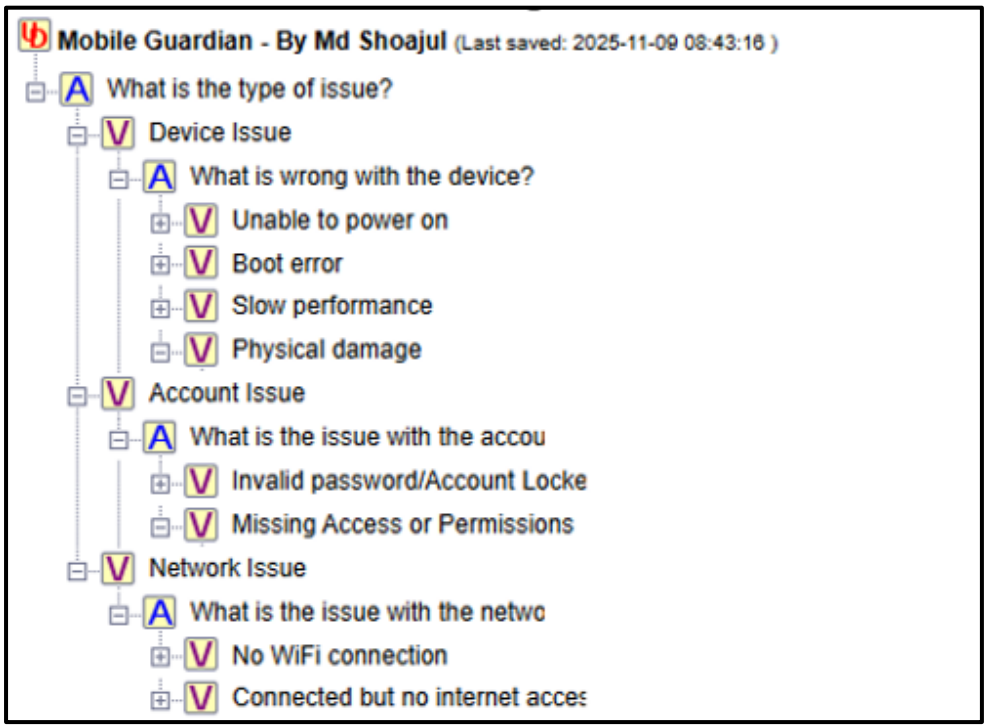
4. Results

At the time of launch, the expert system contains 74 resolutions across device, account, and network issues. Of these, only 24 cases require escalation to the GSA Helpdesk, meaning that approximately 67% of reported cases can be resolved directly by frontline IT support. This highlights the effectiveness of the system in enabling rapid issue resolution without escalation. Previously, all reported issues were logged as tickets and awaited a system expert's response before a technician could perform any resolution, often resulting in delays. By contrast, the expert system allows frontline IT support to resolve the majority of cases directly, significantly reducing turnaround time and improving overall efficiency.

The complexity of resolving issues varies by category, as indicated by both the average number of steps needed to reach a resolution and the proportion of cases that can be resolved directly by frontline IT support.

Issue Category	**Number of Rules**	**Average Steps to Resolution**	**Resolved by Frontline (%)**	**Escalated to GSA Helpdesk (%)**
Device	30	4.7	70%	30%
Account	20	4.9	65%	35%
Network	24	5.7	60%	40%

Network issues generally require the most steps due to layered diagnostics involving connectivity checks, driver updates, and network verification. Account-related issues average 4.9 steps, primarily involving password resets, unlocking, or permissions management. Device issues average 4.7 steps, typically requiring virus removal, battery replacement, or device reimaging. Despite this complexity, the expert system enables frontline IT support to resolve approximately 67% of all cases without escalation, representing a significant improvement over the previous ticket-based workflow.



ES Builder link: [Mobile Guardian Expert System]

5. Future Growth and Adaptation

As the system is used over time, additional cases and unique issues outside the current resolution rules will be logged. These will expand the database of solutions, improving the likelihood that frontline technicians can resolve new cases directly. With continued updates, the percentage of cases resolvable by frontline support is expected to increase, potentially reaching 80% or higher, depending on the complexity of new issues and whether the resolution can be performed by the frontline technician.

Future improvements to the expert system include integration with external systems for automated ticket creation and case tracking, such as the GSA Command Center. The integration could also leverage neural networks and predictive maintenance analytics to automatically classify events and resolve new untriaged cases based on similar past cases.

Additionally, a Machine-Learning Feedback Loop could be implemented using analytics to identify recurrent trends in user interactions and recommend updates to the rules. Knowledge domains could be expanded to cover application-specific problems, and Frequently-Asked-Questions (FAQs) could be added to address related policies, further increasing the system's effectiveness.

6. Discussion

The expert system demonstrates that approximately 67% of cases can be resolved by frontline IT support without escalation, compared to the previous workflow where all issues required ticketing and expert intervention. Network issues required the most steps to resolve, while device issues required the fewest, reflecting the varying complexity across categories. Overall, the system has met its goal of enabling faster, consistent, and efficient issue resolution.

Lessons learned include the importance of maintaining a structured rule set and the value of capturing solutions to unique cases to continually enhance the database. Future enhancements, such as integration with external systems, predictive analytics, and machine-learning feedback loops, are expected to improve resolution rates and operational efficiency further.

In conclusion, this project delivers a feasible and well-rounded design for an expert system to automate technical support at GSA. Implementing this system is expected to lead to significant gains in operational efficiency, faster resolution times for users, and allows human experts to focus on more complex issues. The application of a knowledge system in this domain is not only possible but highly advisable.

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